

Fashion-MNIST Classification

Technical Report

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1. Project Overview

The aim of this project was to build a neural network that can classify Fashion-MNIST images into ten clothing categories. I started with a simple baseline and then changed one main factor at a time. This made it easier to see whether each change actually improved the model or only made it more complicated.

2. Data Preparation

Fashion-MNIST was loaded directly from TensorFlow. The original training data was divided into training and validation sets, while the official test set was kept separate until the final evaluation. Pixel values were converted to float32 and divided by 255, changing the range from 0-255 to 0-1. The class distribution was also checked and found to be balanced.

3. Model Design

The baseline contains a Flatten layer, one Dense layer with 128 ReLU units, and a 10-unit Softmax output. ReLU allows the model to learn non-linear patterns, while Softmax gives a probability for each class. Sparse categorical cross-entropy was used because the labels are integer class numbers. All models used a batch size of 128 and 15 epochs.

4. Controlled Experiments

- **Experiment 1:** Added a 64-unit hidden layer to test whether extra depth improved learning.
- **Experiment 2:** Kept the same architecture and added Batch Normalization and Dropout to improve generalization.
- **Experiment 3:** Lowered Adam's learning rate from 0.001 to 0.0005 without changing the architecture.

5. Results

The table below summarizes the recorded training and validation results. The model was selected using validation results, not test accuracy.

Model	Params	Time (s)	Train Acc.	Val. Acc.	Val. Loss	Gap
Baseline	101,770	42.6	0.9172	0.8878	0.3241	0.0294
Experiment 1	109,386	42.3	0.9224	0.8883	0.3439	0.0341
Experiment 2	109,898	44.5	0.8918	0.8897	0.3086	0.0022
Experiment 3	109,898	45.6	0.8860	0.8823	0.3255	0.0037

Experiment 2 finished with the lowest validation loss. The lower learning rate in Experiment 3 did not improve convergence within 15 epochs; its final validation accuracy dropped to 88.23%.

6. Final Model Performance

Experiment 2 was selected because it had the highest validation accuracy (88.97%), the lowest validation loss (0.3086), and the smallest train-validation gap (0.22 percentage points). On the untouched test set, it achieved 87.68% accuracy, 87.70% macro precision, 87.68% macro recall, and 87.57% macro F1-score. The result was 0.32 percentage points below the required 88% test-accuracy target.

7. Error Analysis

Five incorrect predictions were reviewed in the notebook. Shirt was the most difficult class, with an F1-score of 67.39%. The largest confusions were Coat predicted as Pullover (144 images), Shirt predicted as T-shirt/top (138 images), and Shirt predicted as Pullover (121 images).

These mistakes are reasonable because several upper-body categories have similar outlines, and Fashion-MNIST images are only 28x28 pixels. Details such as sleeves, collars, and fabric shape are sometimes difficult to see.

8. Overfitting and Generalization

The final training-validation accuracy gap was only 0.22 percentage points (89.18% training accuracy versus 88.97% validation accuracy). This shows that overfitting was well controlled. Compared with Experiment 1, Batch Normalization and Dropout reduced the gap from 3.41 to 0.22 percentage points and also reduced validation loss from 0.3439 to 0.3086.

9. Conclusion

The experiments showed that adding one hidden layer slightly improved validation accuracy but increased the generalization gap. Regularization produced the most balanced result, so Experiment 2 was chosen as the final model. Although its 87.68% test accuracy was close to the target, it did not fully reach 88%. If I continued this project, I would test a small convolutional neural network because it can preserve spatial features instead of flattening the image first.